ZHANG Zhenkai

Email: zzk13373667812@gmail.com | GitHub: github.com/ken-ab | Website: ken-ab.github.io/My-personal-website/

EDUCATIONAL BACKGROUND

The Hong Kong Polytechnic University 2026 (Expected)

Incoming MSc Student in Data Science & Analytics

Nanjing Tech University 09/2022-06/2026

B.Sc. in Data Science and Big Data Technology, expected June 2026
Average Score: 88/100

RESEARCH INTERESTS

Large Language Models; Mixture-of-Experts; Efficient Model Adaptation; Generative and Agentic AI

PUBLICATIONS

"Exploring and Enhancing Advanced MoE Models: From DeepSpeed-MoE to DeepSeek-V3." Accepted by the 2025 IEEE 6th International Seminar on Artificial Intelligence, Networking and Information Technology (AINIT 2025). First Author. 03/2025

"Predicting Olympic Medal Performance for 2028: Machine Learning Models and the Impact of Host and Coaching Effect." Applied Sciences, 2025. First Author. 07/2025

"Behavior Prediction of Connections in Eco-Designed Thin-Walled Steel-Ply-Bamboo Structures Based on Machine Learning for Mechanical Properties." Sustainability, 2025. Third Author. 07/2025

"Research Progress on the Integration of Robot Vision, Computer Vision, and Machine Learning: Technological Evolution, Challenges, and Industrial Applications." International Journal of Current Research in Science, Engineering & Technology, 2025. Second Author. 04/2025

"Enhancing Export Potential of Digital Service Trade in BRI Countries: A Stochastic Frontier Gravity Model Analysis of Digital Economy Development and Mediation Pathways." Accepted by DEIC 2025. Third Author. 03/2025

RESEARCH EXPERIENCE

Exploring and Enhancing Advanced MoE Models: From DeepSpeed-MoE to DeepSeek-V3 01/2025-03/2025

Individual Project

- Reviewed the evolution of Mixture-of-Experts architectures and selected seven post-2022 large language models for comparative study.
- Analyzed differences in architectural design and optimization strategies across the selected models.
- Compared MoE and dense Transformer architectures across five dimensions, focusing on performance-efficiency trade-offs.

Predicting Olympic Medal Performance for 2028: Machine Learning Models and the Impact of Host and Coaching Effects 02/2025-07/2025

Group Project

- Categorized countries into four groups based on Olympic participation and medal outcomes, then analyzed expected 2028 medal trends.
- Combined factor analysis, time-series analysis, and four machine learning methods in the MPXG prediction framework.
- Applied an FCNN to estimate the probability that non-medal-winning countries would win their first medal.
- Quantified host-country effects across the previous eight Olympic Games.
- Developed a Bayesian linear-regression scoring framework to estimate coaching effects by discipline.

Behavior Prediction of Connections in Thin-Walled Steel-Ply-Bamboo Structures Based on Machine Learning 07/2024-12/2025

Group Project

- Developed a thin-walled steel-ply-bamboo composite structure combining TWS and bamboo-panel advantages.
- Performed factor analysis and data processing to evaluate connection mechanical properties.
- Compared eight machine learning models for connection classification and identified the strongest-performing approach.
- Applied five hyperparameter-optimization methods to improve model performance and reduce overfitting.

HONORS

Meritorious Winner (M Award), Mathematical Contest in Modeling (MCM) 02/2026

SKILLS

- English: IELTS 6.5 (L 6.5, R 7.0, W 6.0, S 6.0), December 2024
- Programming: Python; scientific computing and visualization with NumPy, Pandas, Matplotlib, and Seaborn
- Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, and K-Means; experience with RNN, LSTM, FCNN, BERT, ChatGLM, and MoE architectures
- Tools and Methods: Git, Linux, regularization, data augmentation, and model evaluation workflows